OSIRIS Offline Pipeline Software User's Manual

Introduction

This document is not meant as a tutorial on how to reduce astronomical data in the optical band but it deals with the OSIRIS Offline Pipeline Software (hereandafter OOPS). The interested reader can find tutorials and manuals (The Zen of IRAF. by A. Charles Pullen. or A User's Guide to CCD Reductions with IRAF by Philip Massey) in the References section.
This document is clearly biased towards IRAF. This choice is due to the simple fact that the authors are IRAF-users. The reader can actually use the program he/she likes best but it will not be covered here. No details will be given about IRAF or any other software related with the OOPS.

The OOPS.py program

Requirements

The OOPS is a python script, therefore python must be installed on the user's computer. The python modules which are used by the pipeline are the following: Tkinter, tkFileDialog, fnmatch, sys, time, os, string, re, math, shutil, pyraf. The absence of one (or more) of these modules will make it impossible to use the pipeline. The installation of these modules is left to the reader. As a quick solution, one could install the SciSoft software collection (www.eso.org/scisoft).
The versions of the program/modules used are the following:

- Python 2.5.1
- PyRAF 1.5
- Numpy 1.0.4
- SExtractor 2.5.0

Moreover, OOPS needs an xgterm for some tasks.
The OOPS has been tested to be working on Mac OS/X (10.5) and Linux (Red Hat).

Downloading and Installing

You can retrieve this software as a tar.gz file on the OSIRIS blog.
Download the file and place it in a directory (e.g. ~/OOPS/ which you will have to create) and unpack the file

```
tar -zxvf .tar.gz
```

This should leave you with:

- a txt file
- an install.sh file
- a lib directory

Install the program:

```
sh install.sh
```

Now you have a python code, which you can launch typing:
./OOPS.py
Note that adding the directory where the `OOPS.py` file is in the environmental variable `$PATH` will lead to unexpected behaviour of the program and it is, therefore, not recommended.
Since the OOPS uses PyRAF, it expects to find IRAF's `login.cl` in the directory `~/iraf/` . Some users have found it better to have the `login.cl` in the same directory where the `OOPS.py` file is.

How the OOPS looks

Once the OOPS is launched from a terminal (from now on we will refer to this terminal as ”the home terminal”), it opens a Graphic User Interface (GUI; see Figure1).

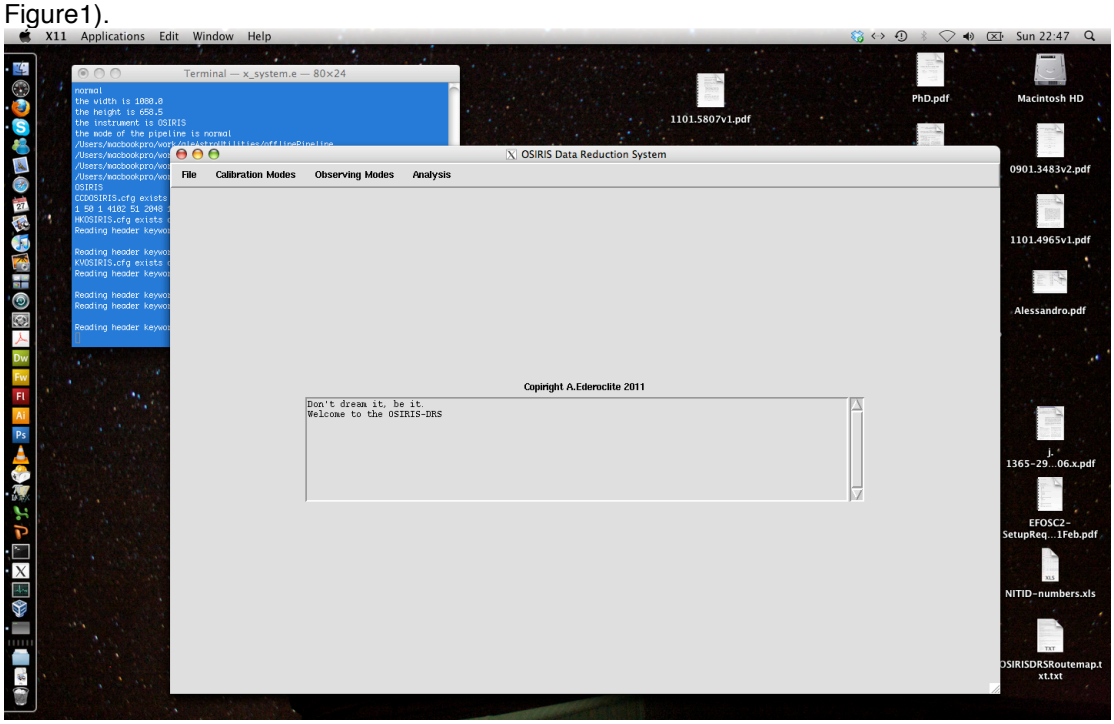


Figure 1: Appearance of OOPS at sartup

At startup, the GUI is organized as follows:

- a menubar
- an empty space which will be later filled with file lists
- a log

The menus are relatively self-explaining and their meaning will become clear during this manual. For completeness, here is a summary of the content of each menu:

- • File
  - Start DS9
  - Select working directories
  - Quit
- • Calibration Modes
  - Imaging
    - Broad Band
      - Bias
      - Darks
      - Dome Flat Fields
      - Sky Flat Fields
      - Super Flat Fields
    - Tunable Filters
      - Standard TF
        - Bias
        - Darks
        - Flat Fields
        - Create Sky Rings
  - Spectroscopy
    - Long Slit
      - Bias
      - Darks
      - Flat Fields

        - Arcs
    - MOS
        - Bias
        - Darks
        - Flat Fields
        - Arcs
- Observing Modes
    - Imaging
        - Broad Band
            - Subtract Overscan
            - Subtract Bias
            - Subtract Dark
            - Divide by Flat Field
            - WCS
            - Make Reduced
        - Tunable Filter
            - Normal Mode
                - Subtract Overscan
                - Subtract Bias
                - Subtract Dark
                - Divide by Flat Field
                - Remove Sky Rings
                - WCS
                - Make Reduced
    - Spectroscopy
        - Long Slit
            - Subtract Overscan
            - Subtract bias
            - Divide by Flat Field
            - Wavelength Calibration
            - Make Reduced
        - MOS
            - Subtract Overscan
            - Subtract bias
            - Divide by Flat Field
            - Wavelength Calibration
            - Make Reduced
- Analysis
    - Imaging
        - Image Quality
    - Spectroscopy
        - Analyse Individual Spectra

When any of the functions is chosen from a menu, the GUI shows (see Figure2):

- the menubar
- three lists of files. Each list showing the fits files contained in three directories described in paragraph 1.4. For each fits file, the contents of the keywords OBJECT, EXPTIME, OUTMODE, RSPEED, GAIN, HBIN, VBIN, OBSMODE, FILTER1, FILTER2, FILTER3, FILTER4, SLITW are shown
- a log

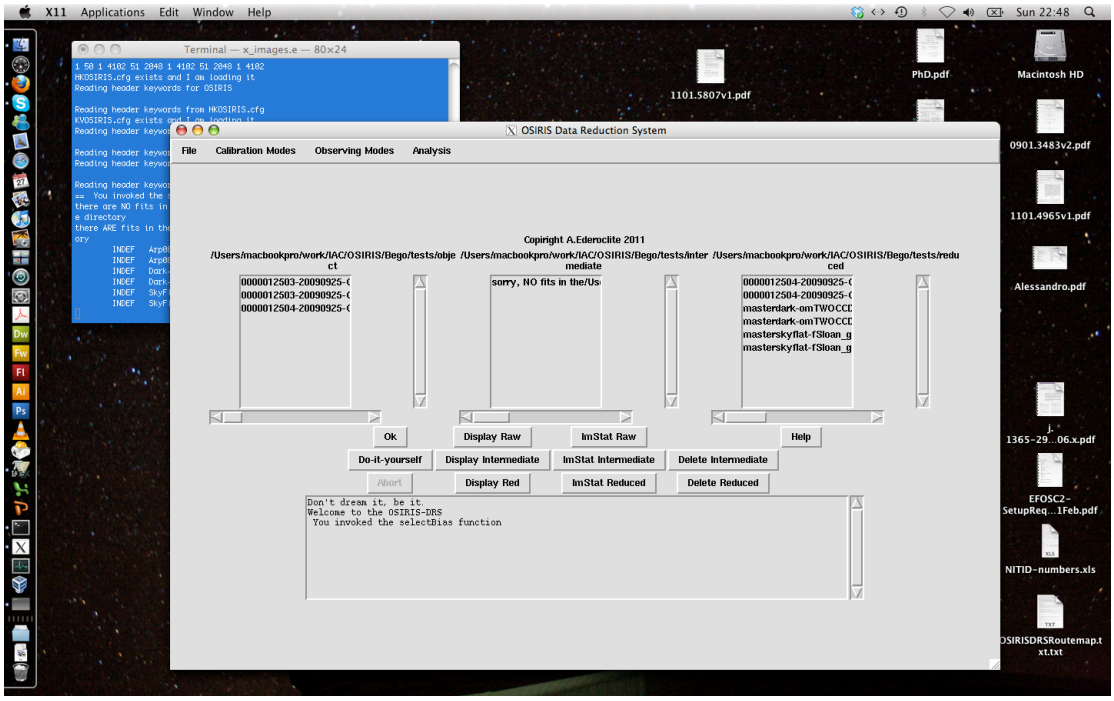


Figure 2: OOPS during execution

How to organize the data

All the data resulting from an observation can be safely located in a directory. We refer to this directory as the ”raw data” directory. When the data are partially reduced (e.g. bias-subtracted but not yet divided by the flat field), they are stored in a directory which is referred to as ”intermediate data” directory. In this directory, each image is split into its extensions. Last, when data are fully reduced, they are stored in a directory called ”reduced data” directory.

The user can decide to put these directories anywhere on his/her computer. The directories can be completely independent from each other, so that a user can have more freedom in organizing his/her data.

The naming of the raw data is responsibility of GranTeCan. The common scheme is <ID>-<date>-OSIRIS-<observing mode>. The intermediate files have the same name as the raw data with added:

- at the end “-ext1” or “-ext2” depending on the extension
- at the beginning:
  - to if the file has been trimmed and overscan subtracted
  - b if the file has been bias corrected
  - d if the file has been dark subtracted
  - f if the file has been flat corrected
  - s if the file has been sky subtracted
  - w if the file has been wavelength calibrated (or the WCS has been applied)
  - p if the file has been photometrically calibrated

The reduced files (including master files) are named as:

- the masterbias is called masterbias and a series of codes referring to the CCD mode
- the wavelength map of a TF image is the same name of the image with appended “wl-map-ext*n*” where “*n*” is the number of the extension
- the master flat of a long slit spectroscopic observation is called “masterspecflat”-<GrismCode>-<CCD status code>-ext*n*.fits (where *n* is the number of the extension)
- the normalized master flat of a long slit spectroscopic observation is called

like the normal master flat but with a trailing “n” (for “normalized”)
- the combined arc is called “masterSpecArc”-<GrismCode>-<CCD status code>-ext*n*.fits (where *n* is the number of the extension)
- the two dimensional arc is called “masterSpecTArc”-<GrismCode>-<CCD status code>-ext*n*.fits (where *n* is the number of the extension)
- the extracted, photometrically calibrated, fully reduced spectrum is called “pwsfbdto”-<fileName>-ext*n*.ms.fits (where *n* is the number of the extension)
- the reconstructed multi-extension fits file of the reduced image, has the same name as the raw image but is located in the “reduced data” directory.

To inform the pipeline where the directories are located, one goes to the menu ”File” ⇒ ”Select working directories”. This opens a pop-up window (pretty intuitive, see Figure 3). In the first entry one should type the directory where the OOPS is located, in the second entry one should type the ”raw data” directory, in the third entry one should type the ”intermediate data” directory and in the fourth entry one should type the ”reduced data” directory. Instead of typing these paths in, one can choose to browse the directory tree using the button on the right hand side of any entry. By clicking on ”OK”, the changes are saved. By clicking on ”Cancel”, the changes are lost. The following picture shows a snapshot of this panel.

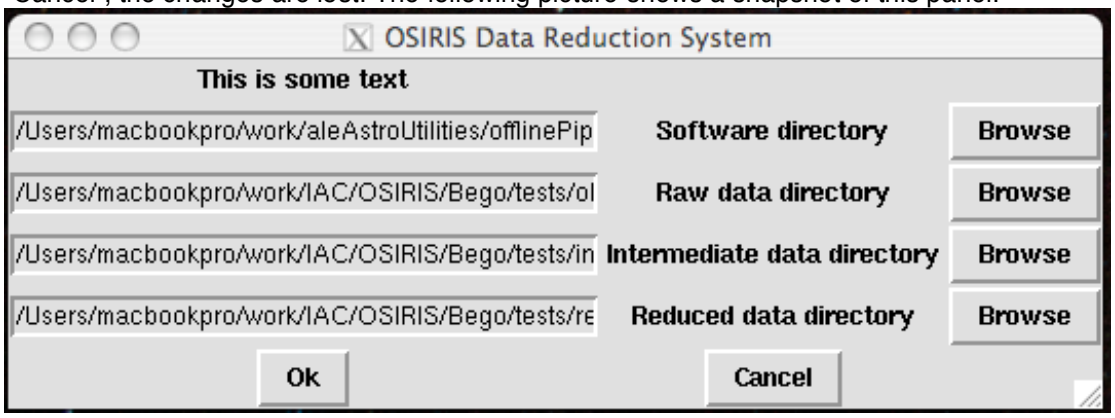


Figure 3: The window for selecting the working directories

General behaviour of the OOPS

It is common, when using IRAF, to keep a DS9 open. Some IRAF tasks, indeed, expect a DS9 to be open. In order to keep all the commands within the OOPS (and under its control), one can (and, indeed, should) open a DS9 from ”File” ⇒ ”Open DS9”. Please, note that the OOPS may encounter problems, if another DS9 is open.
As described above, when selecting a function of the OOPS, one gets three lists of files.
Four columns of buttons are also displayed.

- The first button (”OK”) is needed to execute the selected task automatically.
- The button ”Do-It-Yourself” opens a panel equivalent to the PyRAF ”eparameter” panel to execute the task ”by hand”. This should be done very carefully and, possibly, by experienced users only.
- The button ”Display Raw”, displays in DS9 an image (only one at a time) selected from the ”Raw Data Directory”. The CCD1 is shown in DS9’s frame #1 and the CCD2 is shown in DS9’s frame #2. A good way to see both CCDs is, once displayed, to go to DS9, click on “zoom” and click on “1/4” (this may depend on the size of the monitor being used). Then click on the button “frame”, then click on “tile” (the CCD1 is still at the wrong zoom). Last thing to do is click on “first”, then “zoom” and “1/4”. Both CCDs are now displayed side by side and with the same zoom.
- The button ”Display Intermediate” displays in DS9 an image selected from the ”Intermediate Data Directory”. It is important to stress that, in this case, only one extension is shown (the one chosen) and, therefore, it is displayed in frame #1. To be sure to be looking at the correct frame, in DS9, click on “Frame” -> “First”.
- The button ”Display Red” displays in DS9 an image selected from the ”Reduced Data Directory”. As in the previous case, only one extension is shown and it is shown in the

first frame. Again, to be sure to be looking at the correct image, one should select “Frame” -> “First”.

- The button ”ImStat Raw” prints statistics (Image Name, Number of Pixels, Mean, Standard Deviation, Minimum Value and Maximum Value) of all the extensions of all the selected images from the ”Raw data directory”.
- The button ”ImStat Intermediate” prints statistics (Image Name, Number of Pixels, Mean, Standard Deviation, Minimum Value and Maximum Value) of all the selected images from the ”Intermediate Data Directory”.
- The button ”ImStat Reduced” prints statistics (Image Name, Number of Pixels, Mean, Standard Deviation, Minimum Value and Maximum Value) of all the selected images from the ”Reduced Data Directory”.
- The button ”Delete Intermediate” allows the user to delete (one-by-one) data from the ”Intermediate Data Directory”.
- The button ”Delete Reduced” allows the user to delete (again, one-by- one) data from the ”Reduced Data Directory”. For safety reasons, it is impossible to delete data from the ”Raw Data” directory.
- The button ”Help” shows how the invoked function works.
- Normally OOPS will work selecting files in the “raw data directory”. Files can be selected by pressing the “shift” key and then clicking on the first and last of a group. Otherwise, a user can click on different files keeping the “ctrl” key pressed.

Each time a task is executed by clicking the ”OK” button, the OOPS checks if the execution of this task will overwrite some already existing file and if the execution of this task will require some calibration. In case existing files may be overwritten or needed calibrations are not existing, the OOPS will pop-up a message informing the user of the problem and asking for input (either to delete the existing files and write new ones or stop).
Some tasks may require some time to be executed. A user should be patient and check the messages received in the ”home terminal”. The OOPS hangs up very rarely.

Leaving OOPS

One should never quit the OOPS by clicking on the small cross on the upper right (for Linux users; Mac users have it on the upper left). This ”brute force” approach will avoid the deletion of several temporary files. The proper way to close the OOPS is to click on ”File” ⇒ ”Quit”. This will open a pop-up asking for confirmation. The popup also gives the user the possibility to delete the files stored in the ”Intermediate Data” directory. For safety reasons, the tick mark is always ”off” (i.e. no data will be deleted). If a user wants to delete the data from the ”Intermediate Data Directory”, he/she can just tick the mark.

Preparing Calibrations: Broad Band Imaging

The calibration frames needed for an observation with broad band filters are a bias frame (we will refer to this as masterbias) and a flat field exposure (masterflat).

Combining bias frames

A master bias frame (also known as a ”Zero” image by IRAF users but hereandafter we will call it ”masterbias”) can be produced by going to ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Bias”.
In the ”raw data” list (the file list on the left) one should be able to identify the biases. The user should select the files to be combined from that list and click on “OK”.
To combine bias exposures, they should be taken all with the same readout mode, the same readout speed and the same gain. If a user tries to combine bias frames taken in different ways, the OOPS will refuse and print an error message in the ”home terminal”.
Each of the extensions of the OSIRIS mosaic is subtracted of the overscan value (with the IRAF task ”ccdproc”), it is trimmed (again with the IRAF task ”cddproc”) and they are finally combined (using ”zerocombine” with a ”median”). Overscan and trim regions are hardcoded.

Combining Darks

During the first year of operation, OSIRIS was affected by a dark current problem. If you are reducing data from that period, you will have to subtract a dark. Users reducing later data should not bother since OOPS checks the date and if dark subtraction is required or not.
To get a “masterdark”, go: ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Darks”. In the ”raw data” list (the file list on the left) one should be able to identify the darks. The user should select the files to be combined from that list and click on “OK”.
To combine dark exposures, they should be taken all with the same readout mode, the same readout speed and the same gain. If a user tries to combine dark frames taken in different ways, the OOPS will refuse and print an error message in the ”home terminal”.

Combining dome flat-fields

To produce a dome masterflat (following the same naming scheme used for ”masterbias”), one simply goes to ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Dome Flat Fields”. In the ”raw data” list (the file list on the left) one should be able to identify the flats. The user should select the files to be combined from that list and click on “OK”.
To combine flat fields, images should be obtained, as for bias frames, with the same mode of the CCD. Moreover, all the images should be taken in the same filter. If a user tries to combine images which do not match these characteristics, the pipeline will refuse and print an error message in the ”home terminal”. Each extension of the OSIRIS mosaic is subtracted of the overscan value, it is trimmed, they are subtracted of the ”masterbias” (if any) and they are finally combined (using ”flatcombine” with a ”median”).

Combining sky flat-fields

To produce a sky masterflat (following the same naming scheme used for ”masterbias”), one simply goes to ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Sky Flat Fields”. In the ”raw data” list (the file list on the left) one should be able to identify the flats. The user should select the files to be combined from that list and click on “OK”.
As in the dome flat field case, all images should be taken with the same detector configuration and the same filter. Again, the pipeline will complain if these conditions are not met.
Each extension of the OSIRIS mosaic is subtracted of the overscan value, it is trimmed, they are subtracted of the ”masterbias” and they are finally combined.
If a masterbias is not available, then the pipeline will ask the user whether if he/she wants to stop or if he/she wants to continue, skipping the bias subtraction step.

Combining super flat-fields

To produce a super masterflat (following the same naming scheme used for ”masterbias”), one simply goes to ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Super Flat Fields”. In the ”raw data” list (the file list on the left) one should be able to identify the flats. The user should select the files to be combined from that list and click on “OK”.
As in the dome flat field case, all images should be taken with the same detector configuration and the same filter. Again, the pipeline will complain if these conditions are not met.
Each extension of the OSIRIS mosaic is subtracted of the overscan value, it is trimmed, they are subtracted of the ”masterbias” and they are finally combined.
If a masterbias is not available, then the pipeline will ask the user whether if he/she wants to stop or if he/she wants to continue, skipping the bias subtraction step.

Preparing Calibrations: Tunable Filter Imaging

Combining bias frames

For this, please, refer to the Broad Band Imaging case. Note that you can access this function also from ”Calibration Modes” ”Imaging” ”Tunable Filters ” ”Standard TF” ⇒ ”Bias”.

Combining darks

For this, please, refer to the Broad Band Imaging case. Note that you can access this function also from ”Calibration Modes” ”Imaging” ”Tunable Filters ” ”Standard TF” ⇒ ”Darks”.

Combining dome-flat fields

For this, please, refer to the Broad Band Imaging case . Note that you can access this function also from ”Calibration Modes” ⇒ ”Imaging” ⇒ ”Tunable Filters ” ⇒ ”Standard TF” ⇒ ”Flat Fields”.
If dome-flats have not been obtained for the same tuning of the TF, the OOPS will not combine them.

Preparing Calibrations: Long-slit Spectroscopy

Combining bias frames

For this, please, refer to the Broad Band Imaging case. Note that you can access this function also from ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”Normal Mode ” ⇒ ”Long Slit” ⇒ ”Bias”.

Combining flat frames

For this, please, refer to the Broad Band Imaging case. Note that you can access this function also from ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”Normal Mode ” ⇒ ”Long Slit” ⇒ ”Flats”.
In the ”raw data” list (the file list on the left) one should be able to identify the flats. The user should select the files to be combined from that list and click on “OK”.
Obviously, one can only combine flats which have been taken with the same slit-grism(-filter) combination. In case this condition is not fulfilled, the OOPS will refuse to process the data.
Moreover, the OOPS will fit the flat along the dispersion axis, using the IRAF task “response” and prepare a normalized flat.

Building a wavelength calibration

To calibrate in wavelength, one goes to ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”Normal Mode” ⇒ ”Long Slit” ⇒ ”Arcs”. One then selects the wavelength calibration exposure(s) from the “raw files” list and clicks on “Ok” or “Do-it-yourself”.

Automatic wavelength calibration

The “ok” button allows for automatic wavelength calibration. This means that OOPS tries to fit to the emission lines found in the arcs, a line list (which is available in the “lib” directory). This is based on a series of parameters (stored in the file lib/MOS_displacement.txt) which allow the IRAF task “autoidentify” to make a first guess and then identify automatically the rest of the lines.

Interactive wavelength calibration

The User will have to identify the lines from scratch. It is recommended to use the wavelength calibration atlas on the OSIRIS page in the GTC website.
One should inform the OOPS which lamp has been used. GTC has HgAr, Xe and Ne. It is possible to use either one of the lamps of the three of them.
During the process, the OOPS will open an extra xterm which guide through the process.
An example is given in the following figure:

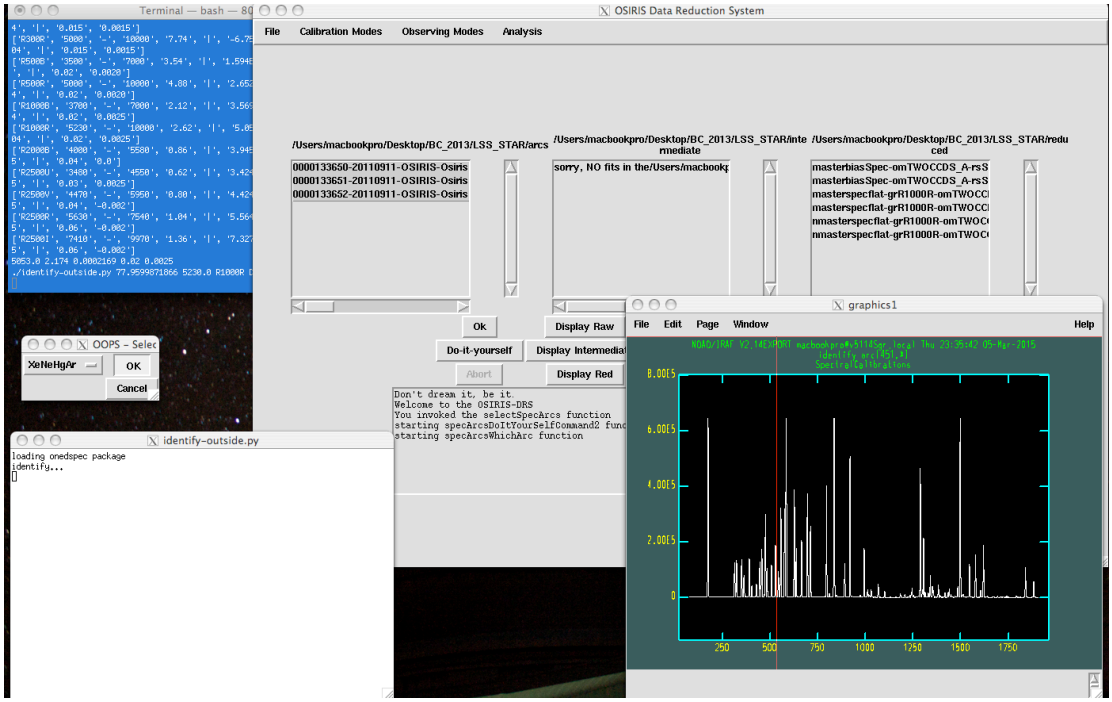


Figure 4: Appearance of OOPS when identifying arcs

The user marks the emission lines typing “m” and the corresponding wavelength (in Angstrom). When a few (at least three) lines are selected, one can attempt a first fit, typing “f”. If the fit is satisfactory (which should), just type “q” to go back to the line identification window and type “l” to get automatically loaded all the lines from a catalogue (the “lib/HgArNeXe-list.txt” file provided with the OOPS) and give, again, “f”.
If the fit is good, then type “q” twice, otherwise type “q” and “i” to restart the process. Also, type “?” for online help.
After this, the OOPS will fit the positions of the lines across the CCD with the “reidentify” function and then map the CCD with “fitcoords”. Finally, as a sanity check, the 2D wavelength calibration is applied to the arc. These images (one per extension) are stored in the reduced directory and named “specTarc”.

Preparing Calibrations: Multi-Object Spectroscopy

Combining bias frames

For this, please, refer to the Broad Band Imaging case. Note that you can access this function also from ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”MOS” ⇒”Normal MOS ” ⇒”Bias”.

Identifying slit-lets

Identification of slit-lets in MOS is done through ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”MOS” ⇒”Normal MOS ” ⇒”Identify Slit Lets”. This is crucial for all the rest of the reduction. The user should select the through-mask image in the “raw files” list. The “through-mask image” is a dome flat-field taken with the mask in the optical path. If such an image is not available, the acquisition image of the field (i.e. the science field observed through the MOS mask) is a fair backup. The “OK” button will use SExtractor to identify the slit-lets while the “Do-It-Yourself” button will allow the User to pick the slitlets interactively.
Note that you need to have SExtractor installed for this.
Also note that you need a DS9 to be running.

Combining flat frames

For this, please, refer to the Broad Band Imaging case. Note that you can access this function

also from ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”MOS” ⇒”Normal MOS ” ⇒”Dome Flat Fields”. The user should select the flat in the “raw files” list. Obviously, one can only combine flats which have been taken with the same slit-grism(-filter) combination. In case this condition is not fulfilled, the OOPS will refuse to process the data.
Moreover, the OOPS will fit the flat along the dispersion axis, using the IRAF task “response”.

Building a wavelength calibration

To calibrate in wavelength, one goes to ”Calibration Modes” ⇒ ”Spectroscopy” ⇒ ”MOS” ⇒”Normal MOS ” ⇒” ”Arcs”. One then selects the wavelength calibration exposure in the “raw files” list and clicks on either “Ok” or “Do-it-yourself”.
As in the case of long slit, if a user selects “Ok”, the OOPS will use the parameters stored in the “lib/MOS_displacement.txt” file to get a wavelength calibration for all the selected slit-lets, taking into account the displacement of the central wavelength of the spectra across the field of view.
In case of selecting “Do-it-yourself”, the User will have to identify the lines from scratch. It is recommended to use the wavelength calibration atlas on the OSIRIS page in the GTC website.
One should inform the OOPS which lamp has been used. GTC has HgAr, Xe and Ne. It is possible to use either one of the lamps of the three of them.
During the process, the OOPS will open an extra xterm which guide through the process, like in the long-slit case (for more details, please, refer to the long-slit scenario).
Note that OOPS will try to display the arcs in DS9 and, therefore, is expecting that DS9 has been launched. Should the pipeline hangs because of this, just start DS9 and restart the wavelength calibration normally.

Reducing Science: Broad Band Imaging

Science exposures reduction is done step by step. The experienced user may find this approach naive or boring. Nevertheless, this gives the user a higher control on the outcome of each step of the reduction process.

Subtracting the overscan

To subtract the overscan, one simply goes to ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Subtract Overscan”. One should select (by clicking)from the raw data list which images he/she wants to subtract of the overscan.
The overscan is hardcoded (and depending on the binning) and it is then subtracted using the ccdproc task in IRAF.

Subtracting the bias

To subtract the masterbias, the user has to select ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Subtract Bias”. One should select (by clicking) from the raw data list which images he/she wants.
As for the calibration steps, the OOPS will check that a proper masterbias exists and that images have been previously subtracted of the overscan. In case images have not been subtracted of the overscan, the user can simply skip the step. In case a proper masterbias does not exist, the OOPS will complain and print a messange in the ”home terminal”.

Subtracting the dark

To subtract the masterdark, the user has to select ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Subtract Dark”. One should select (by clicking) from the raw data list which images he/she wants.
As for the calibration steps, the OOPS will check that a proper masterdark exists and that images have been previously subtracted of the overscan and bias. In case images have not been subtracted of the overscan or of the bias, the user can simply skip the step. In case a

proper masterdark does not exist, the OOPS will complain and print a messange in the ”home terminal”.

Dividing by the flat field

To divide by the flat field, one should select ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Divide Flat Field”. One then selects (by clicking) the file(s) he/she wants to process by selecting them in the “raw data” list. The OOPS will ask which type of flat field the User wants to use. The choices are ”Dome”, ”Sky” or ”Super Flat”.
The OOPS will check for the presence of a proper masterflat (same filter and same detector configuration of the science image(s)). If no proper masterflat is found, the pipeline will complain and give an error message.

WCS

To apply WCS keywords, one can select ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”WCS”. One then selects (by clicking) the file(s) he/she wants to process by selecting them in the “raw data” list.
The OOPS will create a WCS based on the pointing of the telescope, the position angle and the binning. Please, note that this does not replaces in any sense a proper astrometry and it is only meant as a first step.
As in all the other modes, previous calibration is desirable but not strictly required.

Make Reduced

The “make reduced” command, reconstructs the multi-extensions fits file, after reduction, in the reduced directory. Once it is exported like this, one can manipulate it with his/her favourite data analysis software.
This function is accessed from: ”Observing modes” ⇒ ”Imaging” ⇒ ”Broad Band” ⇒ ”Make Reduced”. One then selects (by clicking) the file(s) he/she wants to process by selecting them in the “raw data” list.
Again, it is desirable to go through all the steps of the calibration before making a reduced file but any step can be skipped at any time.

Reducing Science: Tunable Filter Imaging

Subtracting the overscan

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒ ”Tunable Filters” ⇒ ”Standard Mode” ⇒ ”Subtract Overscan”

Subtracting the bias

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒ ”Tunable Filters” ⇒ ”Standard Mode” ⇒ ”Subtract Bias”

Subtracting the dark

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒ ”Tunable Filters” ⇒ ”Standard Mode” ⇒ ”Subtract Dark”

Dividing by the flat field

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒”Tunable Filters” ⇒”Standard Mode” ⇒ ”Divide by Flat Field”.
Please note that the OOPS will only apply a flat-correction if the available flats are for the

same tuning (central wavelength and full width at half maximum) of the science images.

Subtracting the Sky Rings

To subtract sky rings, one access the task at ”Observing modes” ⇒ ”Imaging” ⇒ ”Tunable Filters” ⇒ ”Standard Mode” ⇒ ”Subtract Sky Rings” then one should select the image (for the moment, this command only works on one image at a time) which should be subtracted of the sky ring and then click on ”Ok”. The image is artificially dithered and then combined, thus leaving only the sky ring which is then subtracted from the original image. This method works fine for point-like sources but leaves residuals in case of extended objects.

Preparing a wavelength map

To create an image with a map of the central wavelength corresponding to each pixel of the OSIRIS mosaic, one can use the function “Observing modes” ⇒ ”Imaging” ⇒”Tunable Filters” ⇒”Standard Mode” ⇒ “Wavelength Map”.
This checks the tuning (central wavelength and FWHM) of the science images and uses the program stored in “PhaseField3.py”.
The prepared map can be found in the “reduced data” directory. It has the same name as the science file plus “-wlmap-ext*n*” (where *n* is the number of the extension).

WCS

To apply the WCS, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒”Tunable Filters” ⇒”Standard Mode” ⇒ ”WCS”

Make Reduced

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Imaging” ⇒”Tunable Filters” ⇒”Standard Mode” ⇒ ”Make Reduced”

Reducing Science: Long-slit Spectroscopy

Subtracting the overscan

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒ ”Subtract Overscan”.

Subtracting the bias

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒ ”Subtract bias”.

Subtracting the dark

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒ ”Subtract dark”.

Dividing by the flat field

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒ ”Divide by Flat Field”.

Applying Wavelength Calibration

A dispersion solution should be obtained as described in the calibration section. To apply the solution obtained, one should go to ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒ ”Wavelength calibration”, select the images to be calibrated and click ”Ok”.

Make Reduced

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”Long Slit” ⇒”Make Reduced”

Reducing Science: Multi-Object Spectroscopy

Subtracting the overscan

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒ ”Subtract Overscan”.

Subtracting the bias

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒ ”Subtract bias”.

Subtracting the dark

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒ ”Subtract dark”.

Dividing by the flat field

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒ ”Divide by Flat Field”.

Applying Wavelength Calibration

A dispersion solution should be obtained as described in the calibration section. To apply the solution obtained, one should go to ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒ ”Wavelength calibration”, select the images to be calibrated and click ”Ok”.

Make Reduced

For this, please, refer to the Broad Band Imaging case. Please, note that you can access this by selecting ”Observing modes” ⇒ ”Spectroscopy” ⇒ ”Normal mode” ⇒ ”MOS” ⇒”Make Reduced”

Analysis: Imaging

Check image quality

This function displays the image (one at a time) in DS9 and invokes IRAF task imexamine. The image to be displayed and analyzed should be chosen in the reduced data directory. To call this function, one can go to ”Analysis” ⇒ ”Imaging” ⇒ ”Image Quality”.

Troubleshooting

The OOPS crashes when trying to perform any IRAF related task

This error has been reported by a couple of users already and it is due to the wrong location of the `login.cl` file.
The appropriate location of the `login.cl` file is in a `~/iraf` directory. This is, if your user name is username, the `login.cl` file must be in `/home/username/iraf/` .
It has been reported that also placing the `login.cl` file in the same directory as `OOPS.py` works.

There is an error when trying to copy a file. The file name appears truncated.

This is a feature which has to do partly with the design of the OOPS and partly with IRAF's task "imcopy". The length of the absolute path of a file must be of 135 characters maximum. In case it is impossible to keep within this limit, the User can request the OOPS developer for a small script which renames the OSIRIS files to a shorter file name.

The pipeline hangs when doing wavelength calibration

The pipeline is probably trying to display the images in a DS9 and this has not been started. Just start the DS9 and restart the wavelength calibration.

Memory problem on Linux

A Linux user has reported a problem with memory which was solved with the command:

```
UNIX> ulimit -s unlimited
```

After giving "Cancel" to one of the pop-ups, the pipeline does not say that it is ready

This is a feature that should be fixed soon. The pipeline does not say that it is ready but it is.